

Lagrangian Formulation of K-G Scalar Field and the Electromagnetic Field

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Summary

This note provides a self-contained derivation of the field equations for the Klein-Gordon scalar field and the electromagnetic field from the Lagrangian formulation. The basic framework of the variational principle is first reviewed, followed by detailed computational steps that fill in the intermediate algebra omitted in standard textbook treatments.

Keywords: Lagrangian formulation, variational principle, Klein-Gordon equation, Maxwell equations, field theory

1 Lagrangian Formulation

The Lagrangian formulation can derive the direct tensor formulation from the principle of least action, which not only enhances the elegance of the theory but also provides a bridge toward quantum theory. The basic idea is to construct an action

$$S[\psi] = \int_M \mathcal{L}[\psi] \quad (1)$$

as a functional of the field configuration ψ on the manifold M , where the Lagrangian density

$$\mathcal{L}|_x = \mathcal{L}(\psi(x), \nabla\psi(x), \dots, \nabla^k\psi(x)) \quad (2)$$

is a function of the field ψ and its derivatives up to finite order, with x denoting a point in the manifold M . In the field space, all possible field configurations can be probed by infinitely many one-parameter families ψ_λ . It suffices to require that, at the configuration ψ (corresponding to $\lambda = 0$), the variation of S along any one-parameter family

$$\delta S[\psi] := \left. \frac{dS}{d\lambda} \right|_{\lambda=0} \equiv \int_M \chi \delta\psi, \quad \delta\psi \equiv \left. \frac{d\psi}{d\lambda} \right|_{\lambda=0} \quad (3)$$

exists and always vanishes, i.e.,

$$\chi \equiv \left. \frac{\delta S}{\delta\psi} \right|_{\psi_0} = 0 \quad (4)$$

holds for all such families, thereby singling out the desired field configuration ψ . The procedure is illustrated below with two concrete examples.

2 Klein-Gordon Scalar Field

For the Klein-Gordon scalar field ϕ in *Minkowski* spacetime, the Lagrangian density is given by

$$\mathcal{L}_{\text{KG}} = -\frac{1}{2} \left(\partial_a \phi \partial^a \phi + m^2 \phi^2 \right). \quad (5)$$

Thus $S_{\text{KG}} = \int_M \mathcal{L}_{\text{KG}}$, and consequently

$$\begin{aligned} \left. \frac{dS_{\text{KG}}}{d\lambda} \right|_{\lambda=0} &= \int \left. \frac{d\mathcal{L}_{\text{KG}}}{d\lambda} \right|_{\lambda=0} = -\frac{1}{2} \int \left\{ \frac{d(\partial_a \phi \partial^a \phi)}{d\lambda} + \frac{d(m^2 \phi^2)}{d\lambda} \right\} \Big|_{\lambda=0} \\ &= -\frac{1}{2} \int \left\{ \frac{d[(\partial_a \phi)(\partial^a \phi)]}{d\lambda} + 2m^2 \phi \frac{d\phi}{d\lambda} \right\} \Big|_{\lambda=0} \\ &= -\frac{1}{2} \int \left. \frac{d[(\partial_a \phi)(\partial^a \phi)]}{d\lambda} \right|_{\lambda=0} - m^2 \phi_0 \delta \phi \\ &= -\frac{1}{2} \int \left[\partial_a \phi \frac{d(\partial^a \phi)}{d\lambda} + \partial^a \phi \frac{d(\partial_a \phi)}{d\lambda} \right] \Big|_{\lambda=0} - m^2 \phi_0 \delta \phi. \end{aligned} \quad (6)$$

Since

$$\frac{d(\partial^a \phi)}{d\lambda} = \frac{d(\partial^\mu \phi)}{d\lambda} (e_\mu)^a = \partial^\mu \frac{d\phi}{d\lambda} (e_\mu)^a = \partial^a \frac{d\phi}{d\lambda}, \quad (7)$$

we have

$$\left[\partial_a \phi \frac{d(\partial^a \phi)}{d\lambda} + \partial^a \phi \frac{d(\partial_a \phi)}{d\lambda} \right] \Big|_{\lambda=0} = [\partial_a \phi_0 \partial^a (\delta \phi) + \partial^a \phi_0 \partial_a (\delta \phi)] = 2\partial_a \phi_0 \partial^a (\delta \phi). \quad (8)$$

Hence

$$\begin{aligned} \left. \frac{dS_{\text{KG}}}{d\lambda} \right|_{\lambda=0} &= - \int [\partial_a \phi_0 \partial^a (\delta \phi) + m^2 \phi_0 \delta \phi] \\ &= - \int \partial_a \phi_0 \partial^a (\delta \phi) - \int m^2 \phi_0 \delta \phi \\ &= - \int [\partial_a (\phi_0 \partial^a (\delta \phi)) - \delta \phi \partial_a \partial^a \phi_0] - \int m^2 \phi_0 \delta \phi \\ &= - \underbrace{\int \partial_a (\phi_0 \partial^a (\delta \phi))}_{\text{boundary term, vanishes}} + \int \delta \phi \partial_a \partial^a \phi_0 - \int m^2 \phi_0 \delta \phi \\ &= \int [\partial_a \partial^a \phi_0 - m^2 \phi_0] \delta \phi, \end{aligned} \quad (9)$$

and requiring the above expression to vanish identically yields

$$\frac{\delta S_{\text{KG}}}{\delta \phi} \equiv \partial_a \partial^a \phi - m^2 \phi = 0. \quad (10)$$

This is the desired Klein-Gordon equation.

3 Electromagnetic Field

For the electromagnetic field in *Minkowski* spacetime, the Lagrangian density is given by

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{ab}F^{ab} = -\partial_{[a}A_{b]}\partial^{[a}A^{b]}, \quad (11)$$

where $F_{ab} = \partial_a A_b - \partial_b A_a$. Thus $S_{\text{EM}} = \int_M \mathcal{L}_{\text{EM}}$, and consequently

$$\begin{aligned} \left. \frac{dS_{\text{EM}}}{d\lambda} \right|_{\lambda=0} &= \int \left. \frac{d\mathcal{L}_{\text{EM}}}{d\lambda} \right|_{\lambda=0} = -\frac{1}{4} \int \left. \frac{d(F_{ab}F^{ab})}{d\lambda} \right|_{\lambda=0} \\ &= -\frac{1}{4} \int \left[\frac{dF_{ab}}{d\lambda} F^{ab} + F_{ab} \frac{dF^{ab}}{d\lambda} \right] \Big|_{\lambda=0} \\ &= -\frac{1}{2} \int F^{ab} \frac{dF_{ab}}{d\lambda} \Big|_{\lambda=0}, \end{aligned} \quad (12)$$

where in the last equality we have used $F^{ab} dF_{ab}/d\lambda = F_{ab} dF^{ab}/d\lambda$.

Since $F_{ab} = \partial_a A_b - \partial_b A_a$, we have

$$\frac{dF_{ab}}{d\lambda} = \partial_a \frac{dA_b}{d\lambda} - \partial_b \frac{dA_a}{d\lambda}. \quad (13)$$

Thus, at $\lambda = 0$,

$$\left. \frac{dF_{ab}}{d\lambda} \right|_{\lambda=0} = \partial_a(\delta A_b) - \partial_b(\delta A_a), \quad (14)$$

where $\delta A_a \equiv dA_a/d\lambda|_{\lambda=0}$. Consequently,

$$\begin{aligned} \left. \frac{dS_{\text{EM}}}{d\lambda} \right|_{\lambda=0} &= -\frac{1}{2} \int F_0^{ab} [\partial_a(\delta A_b) - \partial_b(\delta A_a)] \\ &= -\frac{1}{2} \int F_0^{ab} \partial_a(\delta A_b) + \frac{1}{2} \int F_0^{ab} \partial_b(\delta A_a). \end{aligned} \quad (15)$$

Relabeling the indices in the second term ($a \leftrightarrow b$) and using the antisymmetry $F_0^{ba} = -F_0^{ab}$, we obtain

$$\frac{1}{2} \int F_0^{ab} \partial_b(\delta A_a) = \frac{1}{2} \int F_0^{ba} \partial_a(\delta A_b) = -\frac{1}{2} \int F_0^{ab} \partial_a(\delta A_b). \quad (16)$$

Hence the two terms combine to give

$$\begin{aligned} \left. \frac{dS_{\text{EM}}}{d\lambda} \right|_{\lambda=0} &= - \int F_0^{ab} \partial_a(\delta A_b) \\ &= - \underbrace{\int \partial_a (F_0^{ab} \delta A_b)}_{\text{boundary term, vanishes}} + \int \delta A_b \partial_a F_0^{ab} \\ &= \int (\partial_a F_0^{ab}) \delta A_b. \end{aligned} \quad (17)$$

Requiring the above expression to vanish identically for arbitrary δA_b yields

$$\frac{\delta S_{\text{EM}}}{\delta A_b} \equiv \partial_a F^{ab} = 0. \quad (18)$$

Recalling that $F^{ab} = \partial^a A^b - \partial^b A^a$, this can be rewritten as

$$\partial_a \partial^a A^b - \partial^b (\partial_a A^a) = 0. \quad (19)$$

This is the Maxwell equation in vacuum (in Lorentz gauge, $\partial_a A^a = 0$, it reduces to the wave equation $\partial_a \partial^a A^b = 0$).

4 Equations in Curved Spacetime

According to Wald's book [1], for the Klein–Gordon field, the action S_{KG} obtained from the Lagrangian density

$$\mathcal{L}_{\text{KG}} = -\frac{1}{2} \sqrt{-g} \left(g^{ab} \nabla_a \phi \nabla_b \phi + m^2 \phi^2 \right), \quad (20)$$

upon functional differentiation with respect to ϕ , yields the Klein–Gordon equation in curved spacetime. Similarly,

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} \sqrt{-g} g^{ac} g^{bd} F_{ab} F_{cd} = -\sqrt{-g} g^{ac} g^{bd} \nabla_{[a} A_{b]} \nabla_{[c} A_{d]} \quad (21)$$

gives the Maxwell equations in curved spacetime.

References

- [1] Wald, R. M. *General Relativity*. Chicago: The University of Chicago Press, 1984.
- [2] DeepSeek. *DeepSeek* [Large AI Model]. 2024. URL: <https://www.deepseek.com> (Accessed: July 25, 2026).

A Appendix

A.1 Report on Use of AI Tools

I primarily use AI for simple repetitive calculations, as well as for English translation and polishing. AI also assists me in explaining concepts, verifying my ideas, and providing valuable proof strategies.

A.2 What Is New in This Work

This text is my reading notes on Wald's book [1], intended to supplement the specific computational details, while following the basic content of the original work.

Specifically, in Sec. 1, I present the conceptual framework of the Lagrangian formulation based on my own understanding; in Sec. 2, I provide the detailed computational steps leading to Eqs. (E.1.7) and (E.1.8) in Wald's book [Eqs. (6)–(10)]; and in Sec. 3, I give a complete derivation of the equations of motion for the electromagnetic field, starting from the Lagrangian density $\mathcal{L}_{\text{EM}}$ given in Wald's Eq. (E.1.9).